EXECUTIVE SUMMARY

AI-Assisted Relational Database and Sales Analytics Using the Olist E-Commerce Dataset

Individual Submission | October 2026

Module:Deciphering Big Data July 2026

University: University of Essex

Submission Date: 8/10//2026

1. Executive Overview

This project delivers an AI-assisted database analytics solution for an e-commerce organisation using a relational sample derived from the Brazilian Olist public e-commerce dataset. The business requirement is to transform transactional data into a reliable structure that supports sales analysis, management reporting and evidence-based decisions. The submitted sample contains 1,000 delivered orders, 1,000 customers, 1,151 order-item records and 942 products. The relational implementation separates customers, orders, products and order items, allowing transactional facts to be analysed without repeatedly storing customer and product attributes. The wider Olist source contains approximately 100,000 anonymised orders from 2016–2018 and supports analysis across price, freight, customer location and product attributes (Olist, n.d.).

The completed solution combines structured data preparation, SQLite database controls, SQL analytics, AI-assisted Python profiling, feature engineering and an interactive dashboard. AI is treated as a development and interpretation assistant rather than an autonomous decision maker: generated approaches are validated against the source data and database results. This control is important because generative AI can introduce inaccurate or unsupported outputs; NIST (2024) consequently recommends systematic governance, measurement and management of generative-AI risks.

2. Data Requirement, Schema Detection and Data Quality

The client requirement is a dependable analytical view of merchandise performance by product category, customer geography and time. Four source files were therefore profiled before analytical use. AI-assisted Python/Pandas code inspected column names, inferred data types, counted missing values and measured uniqueness. The profiling covered 29 columns. Customers, orders and order_items contained no missing values in the submitted sample. In products, nine records were missing product_category_name and the same nine records were missing product name length, description length and photo quantity, producing 36 missing cells in total. These values were not imputed because replacing unknown descriptive product attributes would create artificial information. Category-level analyses instead exclude null category labels while retaining the underlying transaction where appropriate.

The profiling also identified a semantic type issue: customer_zip_code_prefix is numerically encoded in the CSV but is an identifier rather than a quantity. Treating it as text in future ingestion is preferable because identifiers are not mathematically meaningful and leading zeros may otherwise be lost. Date fields were parsed to datetime for validation and feature extraction, while price and freight_value were retained as numeric measures. This approach follows the principle of data wrangling as a controlled process of cleaning, restructuring and formatting raw data for analysis (Sarkar and Roychowdhury, 2019).

3. Relational Design, Cleaning and Validation

The logical database contains customers, orders, products and order_items. customer_id, order_id and product_id identify the parent entities, while order_items uses the composite key (order_id, order_item_id), reflecting the fact that an order can contain multiple line items. Foreign keys link orders.customer_id to customers.customer_id, order_items.order_id to orders.order_id and order_items.product_id to products.product_id. This normalised structure reduces repeated customer and product information and supports controlled joins for analysis.

Validation was performed at both database and Python levels. SQLite checks confirmed the implemented foreign-key relationships and PRAGMA foreign_key_check returned no violations. SQLite documents foreign keys as constraints that enforce valid parent-child relationships, while

foreign_key_check returns a row for each violation (SQLite, 2026a; SQLite, 2026b). Python validation independently confirmed unique customer and order identifiers, uniqueness of the order-item composite key, complete matching of customer, order and product references, non-negative prices and freight values, and non-negative delivery durations. All nine programmed validation checks passed. Record counts were reconciled to the submitted files, reducing the risk that transformation changed the population.

4. AI-Driven Feature Extraction and Analytical Findings

Feature extraction converted operational fields into business-oriented measures. Merchandise value was defined as item price, consistent with the original project's exclusion of freight from sales. A freight-inclusive value was separately derived as price plus freight_value so the two concepts are not conflated. purchase_month was extracted from order_purchase_timestamp for time-series analysis, delivery_days was calculated from purchase to customer delivery, and product_volume_cm3 was derived from product length, height and width. These features were generated through AI-assisted Python logic and then subjected to deterministic validation rather than accepted solely because they were suggested by AI.

The validated SQL results show merchandise sales of 135,150.38 across 1,000 delivered orders, producing an average order value of 135.15. The leading product category was beleza_saude at 15,187.21, followed by esporte_lazer at 13,440.85 and cama_mesa_banho at 10,877.45. Geographic analysis identified São Paulo (SP) as the strongest state at 51,137.61, followed by Rio de Janeiro (RJ) at 18,358.52 and Minas Gerais (MG) at 12,178.91. November 2017 was the highest-sales month at 12,312.79. These findings indicate concentration by category and geography and a pronounced monthly peak, but they must not be interpreted as proof of market-wide growth because the project uses a selected sample rather than the complete Olist population.

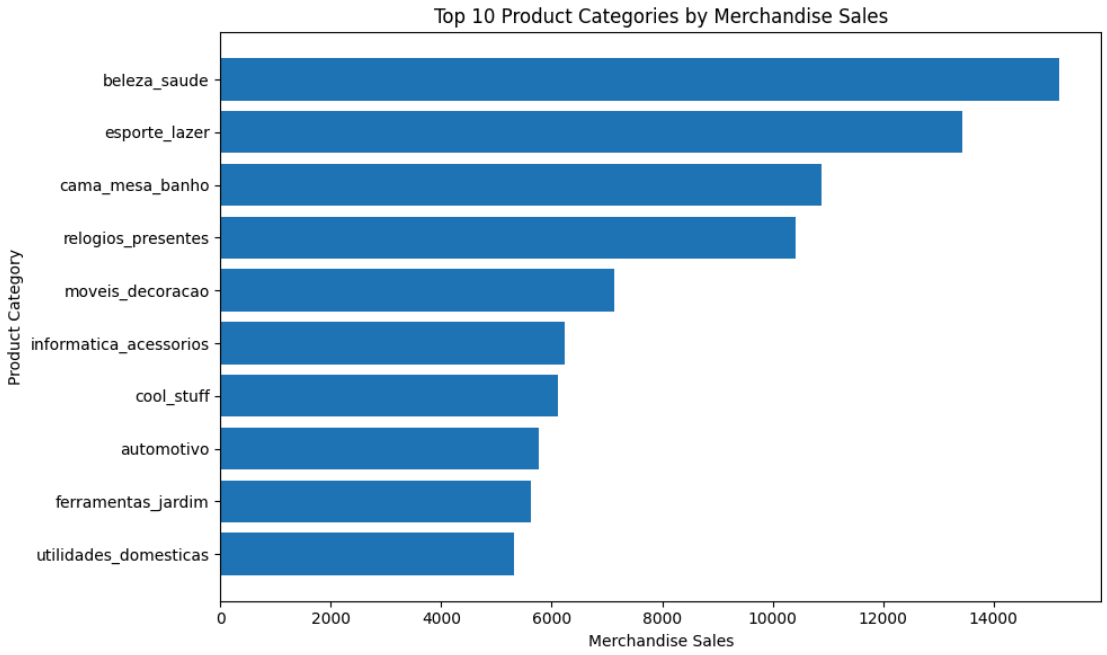


Figure 1. Top 10 product categories by merchandise sales in the selected sample.

5. Intelligent Dashboard and Business Interpretation

An interactive HTML dashboard was produced from the same relational sample to make the validated results accessible to non-technical users. It provides category, customer-state and month preference filters, headline indicators for merchandise sales, record volume and average item value, and linked views for category, geography and monthly trends. The dashboard does not replace the database as the system of record; its purpose is to provide a controlled presentation layer. Filtered results are calculated from the submitted data, and the principal totals were reconciled with the SQL outputs before interpretation.

For management, the most actionable insight is not simply that one category or state is larger, but that commercial attention should be prioritised where value is concentrated while preserving the ability to investigate changes over time. Category filters can support merchandising reviews, state filters can support geographically targeted activity, and month filters can help investigate seasonal or campaign-related peaks. However, the dashboard should be interpreted descriptively. The non-random sample design and exclusion of cancelled or unavailable orders mean that it cannot establish causal effects or represent the entire Olist business.

6. AI Governance, GDPR and Compliance

The dataset is anonymised at source, but pseudonymous customer identifiers and geographic attributes still require careful governance because re-identification risk depends on context and data combination. Under UK GDPR principles, personal data should be processed lawfully and transparently, limited to specified purposes, minimised to what is necessary, kept accurate, retained only as required, protected through appropriate security and supported by accountability controls (ICO, 2026a). In this project, customer identifiers are used to preserve relational integrity and geography is analysed at aggregated state level rather than to profile named individuals. Data minimisation is particularly relevant: the ICO states that personal data should be adequate, relevant and limited to what is necessary for the stated purpose (ICO, 2026b).

Operational deployment would therefore require a documented lawful basis, role-based access, retention rules, audit logging, secure storage and transfer, and procedures for correcting inaccurate personal data. The ICO's accuracy principle requires reasonable steps to ensure personal data is not incorrect or misleading (ICO, 2026c), which reinforces the value of schema checks, referential-integrity controls and reproducible transformations. Integrity and confidentiality also require appropriate security measures (ICO, 2026d). AI-generated code or narrative should not receive uncontrolled access to personal or confidential data; approved environments, human review and traceable validation should be mandatory. NIST's Generative AI Profile similarly frames AI risk management across governance, mapping, measurement and management activities (NIST, 2024).

7. Conclusions and Prioritised Recommendations

The project demonstrates that a small relational e-commerce sample can be transformed into a validated analytical asset. The database design provides clear entity separation and referential integrity; SQL and Python produce consistent sales measures; AI-assisted profiling identifies schema and missing-value issues; engineered features extend the analytical value of the source fields; and the interactive dashboard converts the outputs into a form suitable for business review. The strongest descriptive findings are the concentration of merchandise sales in beleza_saude, the dominance of São Paulo within the sample, and the November 2017 monthly peak.

Priority 1 – Data quality and reproducibility: retain automated schema, missing-value, key and reconciliation checks as a mandatory ingestion gate; store postal-code prefixes as text and document all transformation rules.

Priority 2 – Decision support: use the interactive dashboard for category, state and monthly monitoring, but reconcile KPI logic to SQL and avoid treating sample movements as population-level growth.

Priority 3 – Analytical coverage: repeat the analysis on a larger, preferably reproducible random sample or the full dataset and incorporate additional dimensions such as sellers, payments and delivery performance where business requirements justify them.

Priority 4 – Governance and compliance: apply purpose limitation, data minimisation, retention, access control and auditability; use only institutionally approved AI environments for data that could identify or profile individuals.

Priority 5 – AI assurance: retain human review of generated code and narrative, record prompts and assumptions where appropriate, and validate every material AI-derived insight against deterministic database or Python outputs.

References

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Appendix A. Validation Evidence

AI-assisted Python profiling identified 29 source columns and 36 missing cells, all within four descriptive product fields across nine product records. The generated schema/missing-value profile and feature-validation file are supplied separately as supporting evidence. The interactive dashboard is also supplied as a separate HTML file.

Validation check	Result
Customer IDs unique	PASS
Order IDs unique	PASS
Order-item composite key unique	PASS
All order customer IDs matched	PASS
All item order IDs matched	PASS
All item product IDs matched	PASS
Prices non-negative	PASS
Freight values non-negative	PASS
Delivery days non-negative	PASS